

Adaptive Unsupervised Anomaly Detection for Low-Quality Multivariate Time-Series Data

Jiayi Liu, Donghua Yang, Jinbao Wang, Hong Gao, Jianzhong Li

Abstract—How should we perform anomaly detection on multivariate time-series data with missing data, attribute misplacement and concept drift? The majority of existing anomaly detection methods overlook the fact that data are often of low quality. To address this challenge, we propose an adaptive unsupervised anomaly detection method for low-quality multivariate time series data. Our method introduces a self-attention mechanism that integrates masked information and missing length information to enhance the model's capability in handling incomplete data. Furthermore, we design a deep probabilistic adaptive memory network to improve the model's adaptability to attribute misplacement and concept drift. We also discuss the optimal window size for effectively dealing with concept drift. Comparative experiments on multiple real-world datasets demonstrate that our method can effectively detect anomalies in low-quality multivariate time series data. The experimental results further highlight the robustness of our model, proving its ability to maintain high performance in the presence of data quality issues.

Index Terms—Anomaly detection, Low-Quality, Multivariate Time-series data.

I. INTRODUCTION

Multivariate time series are widely exist in industries, finance, and the internet, among other fields. By conducting in-depth analysis of multivariate time series data, we can reveal hidden abnormal patterns and behaviors within the data. For instance, in the industrial field, analyzing these data helps us monitor equipment performance and prevent potential failures. In the financial industry, in-depth analysis of this data can help us identify abnormal transactions and prevent fraudulent behavior. In the field of cyber security, analyzing these data is crucial for uncovering attack patterns and ensuring the security and stability of networks. Anomaly detection, as a key means of identifying and warning of these potential risks, provides strong support for risk management and decision-making [1]. Therefore, the development of accurate and robust anomaly detection for multivariate time series is particularly important.

Due to the lack of labels in the collected data and the high cost of manual annotation on large-scale datasets, the advantages of unsupervised learning methods in anomaly

detection of multivariate time series data are highlighted. However, traditional unsupervised methods, such as distance-based methods [2], density-based [3], and tree-based methods [4], have some limitations in capturing the complex spatio-temporal dependencies within multivariate time series and in handling large-scale data [5]. In recent years, neural networks have become a research hotspot for deep learning based multivariate time series anomaly detection methods due to their powerful feature learning and representation capabilities [6]. For instance, autoencoders, as a powerful unsupervised learning model, can effectively cope with the impact of noise and identify anomalies by comparing the differences between input and reconstructed output, thus being widely used.

Deep learning-based unsupervised anomaly detection aims to learn general patterns from normal data to enable accurate anomaly detection. Although existing methods have achieved strong performances, most of them rely on well-structured time series data while overlooking the pervasive data quality issues in real-world scenarios. In practice, time series data in real-world settings often suffer from missing data and attribute misplacement caused by errors in data transmission or acquisition, as well as distribution drift due to dynamic data changes. These issues severely impair a model's ability to learn and reason, thereby posing fundamental challenges to building high-precision and robust anomaly detection systems. Specifically, existing methods still face significant challenges in the following three aspects.

Firstly, in the real world, missing data is a common issue. Data can be lost during the collection process due to equipment failures, environmental factors such as sensors being affected by weather or electromagnetic interference, transmission errors such as data transmission interruptions caused by system maintenance or updates, and human errors such as medical testing equipment not being properly connected to patients [7], [8]. However, missing values can disrupt the continuity and integrity of the data, which may lead to inadequate model training and reduced detection quality. It can be seen that the existence of missing values fundamentally affects the performance of anomaly detection algorithms [9], [10]. Existing methods mainly focus on imputing missing data reasonably [5], [10]. However, these methods may mask anomalies and introduce errors. Additionally, they may increase computational complexity due to sophisticated imputation methods.

Secondly, during the data collection process, misplaced attribute values may occur, where a value of one attribute is erroneously entered into another attribute, resulting in incorrect swaps between attribute values. Misplaced attribute values are often observed in practice, such as in IoT environments

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where workers might accidentally plug the sensor cables of a wind turbine in the wrong positions, causing a misplacement in the recorded data for attributes like voltage and temperature [11]. When integrating data from different sources, the process of information extraction and transformation can also easily lead to misplaced attribute values [12]. Misplaced attribute values are a type of error rather than an anomalies because the correct data is indeed existing in the dataset, just placed in the wrong location [11]. Typically, anomalies are defined as observations that deviate from the normal pattern [13], [14]. Unfortunately, misplaced attribute values also exhibit this behavior, leading to the possibility that existing anomaly detection may misidentify them as anomalies, which can cause significant bias in data analysis and decision-making. Therefore, the presence of misplaced attribute values in time series significantly reduces the accuracy of anomaly detection. However, the issue of misplaced attribute has not yet been addressed in existing research [11].

Thirdly, the dynamic nature and non-stationarity of real-world environments cause the data distribution to change over time, a phenomenon known as concept drift [15], [16]. The presence of concept drift can affect the accuracy of anomaly detection [17], [18]. Since deep anomaly detection models are typically trained on historical data, when the data distribution changes during anomaly detection, the model may not effectively adapt to the new data distribution, thereby affecting its performance to detect anomalies. To ensure the accuracy of anomaly detection methods, it is necessary to identify and adapt to changes in data distribution.

In recent years, several methods have been proposed for anomaly detection in time series with missing data, attribute misplacement or concept drift [5], [19], [20]. However, these researches have not consider the above three challenges at the same time. However, these methods can only address one specific challenge individually and are unable to tackle the three major issues in low-quality data: missing data, attribute misplacement, and concept drift at the same time.

To address the aforementioned challenges, this paper proposes an Adaptive Unsupervised Anomaly Detection method (AUAD). It consists of two major networks: the Self-Attention and Convolutional Neural Network(SA-CNN) and the Deep Probabilistic Adaptive Memory Network.

In the SA-CNN network, we introduce a self-attention mechanism that incorporates masking and missing length information. This mechanism provides the model with temporal contextual information about missing values, enabling it to handle missing data more flexibly. Subsequently, we employ a multi-scale convolutional neural network for encoding, allowing the model to understand the integrity and structure of the data at different scales. This effectively captures the spatiotemporal features of the time series. We further introduce a memory network to enhance the model's ability to remember normal data patterns. Specifically, the memory module is designed to store normal feature representations learned from the probabilistic model. By reintegrating the encoded features of the input data with the similarity-weighted items from the memory module, the model's adaptability to misplaced attribute is significantly enhanced.

The feature representations obtained from the two modules are then fused into a final representation for reconstruction. The two modules work in concert, leveraging the feature extraction and memory augmentation mechanisms to effectively address anomaly detection in low-quality data.

During the detection phase, the anomaly score is computed by combining the similarity between the input data and the memory module with the reconstruction error. To adapt to changes in the data distribution, we dynamically update the memory information and threshold of the memory network based on extreme value theory. This approach enhances the accuracy and robustness of anomaly detection. Additionally, we theoretically demonstrate the optimal window size for effectively handling concept drift.

The contributions of this paper are highlighted as below.

1. To our knowledge, this paper is the first study to propose anomaly detection for multivariate time series data in the presence of missing data, attribute misplacement, and concept drift. To this end, we have designed an adaptive unsupervised anomaly detection method to address the challenge of detecting anomalies in low-quality multivariate time-series data.
2. This paper proposes a self-attention mechanism that combines masking and missing length information to enhance the model's ability to handle incomplete data.
3. This paper proposes a method that enhances the model's ability to handle misplaced attribute by re-integrating the encoded features through fusion with similarity-weighted items from the memory module.
4. This paper proposes a method for dynamically updating the memory module, enabling the model to adapt to concept drift in time-series data. Moreover, we discuss the optimal window size for effectively handling concept drift.
5. Experimental results demonstrate that our method achieves high accuracy and robustness in the presence of low-quality time-series data.

II. RELATED WORK

In recent years, deep learning-based anomaly detection have been proposed and demonstrated well performance [21]. Compared to traditional anomaly detection algorithms, their main advantage lies in their strong ability to learn nonlinear features. Currently, unsupervised multivariate time series anomaly detection methods can be mainly divided into two categories: prediction-based models and reconstruction-based models.

The reconstruction-based models are to detect anomalies through reconstruction errors. USAD [22] proposed a reconstruction method based on adversarial learning, which trains two AE adversarially to amplify the reconstruction errors of inputs containing anomalies. MADGAN [23] uses LSTM-RNN as the basic model for GAN generators and discriminators to capture dependencies in time series data and integrates reconstruction errors and discrimination errors as the basis for judging anomalies. MSCRED [24] proposes an attention-based convolutional long short-term memory (ConvLSTM) network to capture spatiotemporal dependencies and utilizes a convolutional decoder to reconstruct the feature matrix for end-to-end

learning, OmniAnomaly [25] combines the advantages of GRU and VAE, learning the normal patterns of multivariate time series through a stochastic recurrent neural network model, and uses reconstruction probabilities for anomaly judgment. InterFusion [26] models the interdependencies and internal dependencies of multiple time series through a hierarchical Variational Auto-Encoder with two random latent variables, and finally detects anomalies based on reconstruction information. The EdgeConvFormer model [50] integrates dynamic graph convolution with Transformer, employing Time2Vec encoding and parallel sensor-specific attention to extract spatiotemporal features, and achieves anomaly detection in multivariate time series via reconstruction error. Liu et al. [51] further introduce a novel "localization score" combined with physics-informed knowledge, enabling EdgeConvFormer to perform both anomaly detection and localization, thereby enhancing the model's interpretability. CT-DDPM [52] proposed a denoising diffusion probabilistic model that integrates a Gaussian Copula with a Transformer architecture. This method models the joint distribution of noise using a Gaussian Copula during the forward diffusion process and reconstructs the normal data distribution through the reverse denoising process, thereby enabling effective anomaly detection in multivariate time series.

Based on prediction models detect anomalies by predicting the future values of time series. Hundman et al. [27] proposed an improved LSTM model that proposes a non-parametric dynamic threshold to assess prediction errors. GDN [28] combines sensor embedding, graph structure learning, and graph attention prediction, effectively learning the relationship graph of sensors from multivariate time series data and detecting anomalies within it. GTA [29] combines multi-scale dilated convolution and graph convolution to obtain encodings, which are then fed into the Transformer model to build and predict time series data for future moments. RC-Attention [53] proposed a model that combines empirical mode decomposition with recurrent convolutional encoding attention. It captures the chaotic temporal dependencies of sequences through phase space reconstruction and recurrent convolutional encoding, reduces the interference of anomalous states on prediction using a centered steady attention mechanism, and finally identifies anomalies based on prediction errors.

Combining prediction-based and reconstruction-based models, such as MTAD-GAT [30] utilizes two parallel graph attention network layers to learn the temporal and feature dependencies among multiple time series, CAE-M [31] model combines a deep convolutional autoencoder with an attention-based bidirectional LSTM, aiming to describe complex spatiotemporal patterns through simultaneous reconstruction and predictive analysis. DAGMM [32] combines a reconstruction-based autoencoder with a probability density-based GMM to enhance the accuracy of anomaly detection. Xu et al. [33] proposed a model based on an association-based criterion and reconstruction, capturing the reconstruction characteristics of data indirectly by evaluating the differences in association between data points and sequences, thereby achieving anomaly detection.

The aforementioned researches have overlooked a fact that, in addition to noise and anomalies in the data, there are

also issues such as missing data and attribute misplacement. Regrettably, attribute misplacement has not yet been explored [11]. Regarding the issue of missing data, Xu et al. [9] proposed an unsupervised anomaly detection algorithm called Donut based on the Variational Auto-Encoder (VAE), which incorporates missing data into the training to enhance the model's robustness against anomalous data. FluxEV [10] applies preprocessing on missing data and designs two interpolation strategies. GST-Pro [5] proposed dynamic graph neural differential equations (DG-NCDEs) to model multivariate time series, effectively capturing the spatiotemporal features of the data even when it contains missing values. Based on a deep probabilistic graphical model, SCNF-EM [34] is designed as a new framework for unified data imputation and unsupervised learning to improve the robustness of anomaly detection in the presence of missing data. AD-CIFC [8] is an anomaly detection method that combines data imputation and feature collaboration. Besides, it introduces a new method for calculating the loss function under missing data conditions.

In addition, in dynamic environments, the distribution of data changes over time, which is known as concept drift. However, static detection methods often have a decrease in accuracy over time and environmental changes due to their lack of adaptability. Siffer et al. [17] proposed the extreme value theory-based anomaly detection SPOT and DSPOT, which are only sensitive to extreme values and have limited adaptability to concept drift. StepWise [18] proposed a framework that can let any type of anomaly detection algorithm quickly adapt to concept drift. However, this method can only handle the data that trends are almost consistent before and after the occurrence of concept drift. Based on RNN for multi-step prediction of time series, Saurav et al. [19] use prediction errors to update the model and enable the model to adapt to changes in data distribution.

In summary, there is currently no research that discusses anomaly detection in the presence of missing data, attribute misplacement, and concept drift. The limitations of existing methods have inspired our research on anomaly detection in multivariate time series when facing these challenges at the same time.

III. PROBLEM FORMULATION

A. Problem Definition

Definition 1(Multivariate time series) A multivariate time series can be represented as $X = (X_1, \dots, X_n)^T \in \mathbb{R}^{n \times T}$, where n represents the total number of attributes, and T denotes the data length of each attributes. The value of the i^{th} attribute at j^{th} timestep is denoted as X_{ij} .

Anomaly detection: To take into account contextual information, we apply a sliding window of length W on multivariate time series (MTS) to calculate the anomaly detection results. Specifically, we calculate the detection score for the sample data within the current window; if it exceeds the model's preset threshold, the sample is marked as an anomaly.

B. Overview

In this paper, we propose an Adaptive Unsupervised Anomaly Detection(AUAD) method for low-quality data,

which consists of two main components, as illustrated in Fig.1. Fig.1(a) is an attention-based multi-scale convolutional network, and Fig.1(b) is a deep probabilistic adaptive memory network.

In the SA-CNN network, the input data X is first concatenated with the missing mask M and temporal information T to form an enhanced data representation. This step aims to integrate crucial temporal and missingness information to enrich the semantic content of the data. Subsequently, the data is re-represented through a self-attention mechanism to improve data integrity and feature expressiveness. The re-represented data is mapped into a latent feature space via a multi-scale convolutional neural network to further extract and understand the complex spatiotemporal features of the data. In the memory network, the input data X is transformed into a feature representation Z through an RNN network. The memory module stores feature information derived from Z after probabilistic distribution modeling.

The feature representation Z is re-integrated by fusing it with the similarity-weighted items of the memory module to obtain a new encoded feature Z_m . Subsequently, Z_m is combined with the features Z_c extracted by the convolutional network and passed through a fully connected layer for reconstruction. Finally, an integrated anomaly scoring mechanism is constructed by combining the similarity between the data and the memory module with the reconstruction error of the data. This mechanism employs extreme value theory to set the critical threshold for anomaly detection, which is used to determine whether a data point is anomalous and whether the information stored in the memory module needs to be updated to adapt to new changes in the data distribution. This approach not only enhances the accuracy of anomaly detection but also enables the memory module to dynamically adapt to the evolution of the data, thereby improving the model's adaptability and robustness for time-series anomaly detection.

IV. METHOD

A. Self Attention-based Multi-Scale Convolutional Neural Network

The key to anomaly detection in multivariate time-series data lies in the ability to capture temporal dependencies within variables and correlations between variables. In order to capture the spatiotemporal features of data more accurately and reliably, we introduce missing position and length information to comprehensively consider the missing data regions.

We designed a mask matrix M and a missing length information T to assist in the encoding of raw data and the acquisition of latent space. The primary function of the mask matrix M is to indicate which data are missing. The missing length information T , on the other hand, can mark the duration of the missing data. The construction formulas for the mask matrix M and the missing length information T are as follows,

$$M_{ij} = \begin{cases} 1 & \text{if } X_{ij} \text{ is observed} \\ 0 & \text{if } X_{ij} \text{ is missing} \end{cases}, \quad (1)$$

$$T_{ij} = \begin{cases} 1 + T_{i(j-1)} & \text{if } M_{ij} = 0 \\ 0 & \text{if } M_{ij} = 1 \end{cases}, \quad (2)$$

where X_{ij} represents the data corresponding to the j^{th} timestamp of the i^{th} attribute. If $M_{ij} = 0$, it means that data X_{ij} is missing. For any i^{th} attribute in j^{th} timestamp, if data X_{ij} is missing, The value of T_{ij} is obtained by adding 1 to the value from the previous moment.

Given a multivariate time series data X , along with the mask matrix M and the missing length information T , we construct a new input matrix $\hat{X}$ as follows:

$$\hat{X} = \text{Concat}(X, M, T), \quad (3)$$

where Concat is a concatenation operation that can combine the temporal data X , the mask matrix M , and the missing length information T to form a new matrix.

To enhance the accuracy of the model's reconstruction results, we attempt to artificially create missing data for the existing data X . For each batch of data input into the model, we artificially make some missing of the originally existing data at a certain ratio randomly. These values do exist, but they are invisible to the model, i.e., artificially missing. To calculate the reconstruction of these artificially missing values in the subsequent loss function, we construct an artificial missing indicator matrix I as follows.

$$I_{ij} = \begin{cases} 1 & \text{if } X_{ij} \text{ is artificially masked} \\ 0 & \text{otherwise} \end{cases}, \quad (4)$$

where X_{ij} represents the data corresponding to the j^{th} timestamp of the i^{th} attribute. If $I_{ij} = 1$, it means that data X_{ij} is an artificial missing. It is important to note that regardless of whether the data is naturally missing or artificially missing, our M matrix will mark it out, while the I matrix will only mark the artificially missing data.

In the field of time series data research, the attention mechanism is also an effective way to focus on important data and information. We use self-attention mechanisms to mine the effective information from time series data. Attention weights are calculated through matrix dot products and softmax functions. Given the query matrix Q , key matrix K , the matrix $\hat{X}$ get by formula (3) and value matrix V as follows,

$$Q = W_q \hat{X}, K = W_k \hat{X}, V = W_v \hat{X}, \quad (5)$$

where $W_q \in \mathbb{R}^{d_k \times d_k}$, $W_k \in \mathbb{R}^{d_k \times d_k}$ and $W_v \in \mathbb{R}^{d_k \times d_k}$ are the attention weight matrices for the query matrix Q , key matrix K , and value matrix V , respectively. The d_k is the dimension of weight matrices. The formula for the self-attention mechanism is as follows,

$$\text{SelfAtt}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

We take $\hat{X} = W_v \hat{X}$, with the added attention weights, as the input to the multi-scale convolutional network.

We constructed a multi-scale convolutional network. Multi-scale convolutional neural networks bring different receptive fields to the network without introducing too many parameters and computational complexity. We uses convolutional layers to gradually reduce the size of the input data and extract feature representations of the latent space. Similarly, the fully connected networks gradually restores these features to their original size through upsampling operations. The training

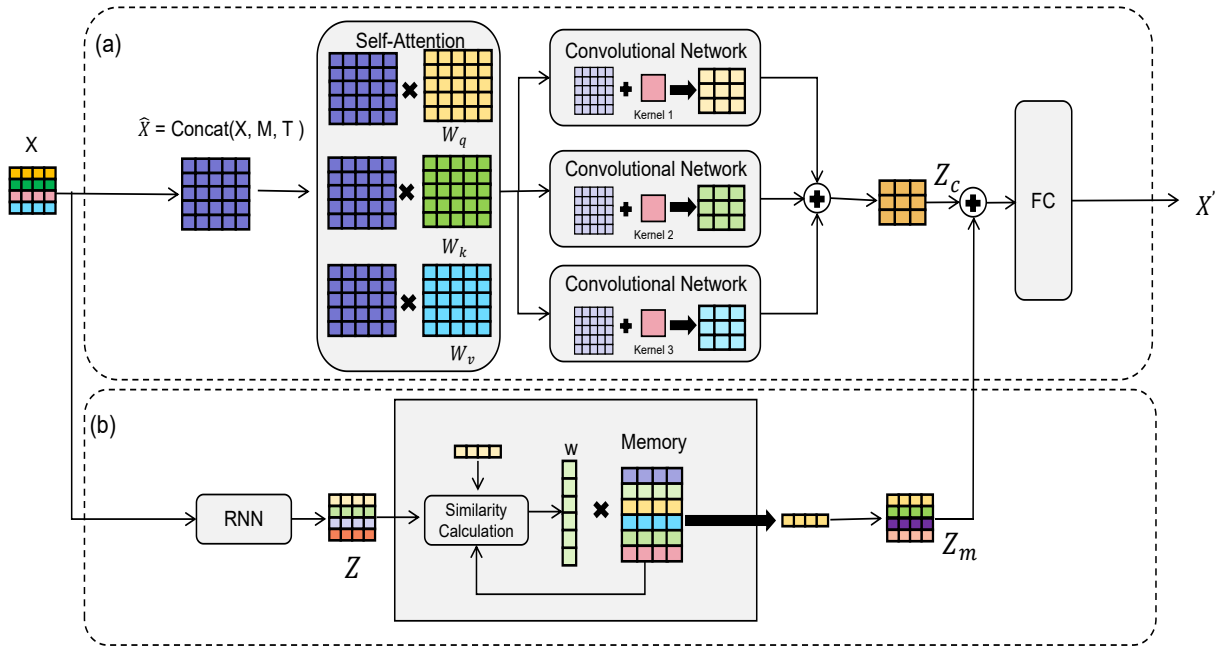


Fig. 1. The overview of unsupervised anomaly detection for low-quality data.

objective of the network is to reduce the reconstruction error between the input data and the output data. In this process, we capture the temporal and spatial relationships between different attributes. Secondly, based on the mask matrix M and the missing length information T , we effectively obtain the latent representations of the data.

Specifically, assuming there are m different scales of convolution, which means there are m convolutional kernels. Each convolutional kernel is composed of a set of parameters $\mathbf{W}_m$ and $\mathbf{b}_m$ then the output of the convolution can be expressed as follows.

$$\mathbf{c}_m = \text{Conv2D}(\hat{\mathbf{X}}\mathbf{W}_m + \mathbf{b}_m), \quad (7)$$

$$\mathbf{h}_m = f_1(\mathbf{c}_m), \quad (8)$$

where Conv2D is the two-dimensional convolution and f_1 is a linear function. The convolution layer includes three convolutional kernels with sizes of 3×3 , 5×5 , and 7×7 respectively. The stride for all is 2. No activation functions are used in the process of reconstructing the data.

This means that according to formula (7) and (8), we can obtain $\mathbf{h}_1$, $\mathbf{h}_2$, and $\mathbf{h}_3$ which are affected at different convolutional scales. By combining the hidden layers at different scales, we get the latent space $\mathbf{Z}_1$ as follows,

$$\mathbf{Z}_1 = (\mathbf{h}_1 + \mathbf{h}_2 + \mathbf{h}_3)/3. \quad (9)$$

After the multi-scale convolutional network performs latent feature extraction on the input data, the output $\mathbf{Z}_1$ is connected to a multi-layer fully connected network, and then a Flatten layer is used to flatten the obtained latent feature representation into 1D data, denoted as $\mathbf{Z}_c$. The deep probabilistic adaptive memory network for analyzing attribute misplacement and concept drift will obtain a latent representation of the data distribution, which is flattened and denoted as $\mathbf{Z}_m$ (details are

described in Subsection 4.2). Combining the two latent space in a comprehensive latent space $\bar{\mathbf{Z}}$ as follows,

$$\bar{\mathbf{Z}} = \alpha\mathbf{Z}_c + (1 - \alpha)\mathbf{Z}_m, \quad (10)$$

where α is a weight parameter used to measure the importance of the two latent space representations.

Then, we use a multi-layer fully connected network to gradually restore these latent space representations into reconstructed data $\mathbf{X}'$. The formal representation is as follows,

$$\mathbf{X}' = FC(\bar{\mathbf{Z}}\mathbf{W}_d + \mathbf{b}_d), \quad (11)$$

where FC represents the multi-layer fully connected network. $\mathbf{W}_d$ and $\mathbf{b}_d$ are the weight parameter vector and bias vector of the fully connected layer, respectively.

B. Deep Probabilistic Adaptive Memory Network

Inspired by trajectory anomaly detection [35], we can analogize attribute misplacement to trajectory switching. From this perspective, each attribute can be regarded as an independent trajectory, and attribute misplacement is equivalent to the process of switching from the current trajectory to another. Unlike trajectory anomaly detection, our goal is not to identify attribute misplacement itself, but to reduce the false positives caused by attribute misplacement. To address this issue, we designed a memory module that stores the time-related encoded features of each attribute. Then, by querying the similarity with memory items, we integrate the encoded features to eliminate the bias in the input data caused by attribute misplacement. We will introduce a deep probabilistic memory module and provide a detailed description of the adaptive memory module below.

First, we construct an RNN model to map data into vector representations in a low-dimensional space. The RNN structure can effectively capture information along the timeline. At each time step t , the RNN updates its hidden state based on the current input x_t and the hidden state of the previous time step h_{t-1} :

$$h_t = f_2(x_t, h_{t-1}), \quad (12)$$

where f_2 is a deterministic non-linear function to be learned, which can be implemented with LSTM or GRU.

Due to the presence of noise and missing data in the input, to address the impact of this uncertainty, we use probability distributions to obtain latent features.

$$q_\phi(z|x) = \mathcal{N}(\mu_x, \text{diag}(\sigma_x^2)), \quad (13)$$

where $\mu_x \in \mathbb{R}^M$ and $\sigma_x \in \mathbb{R}^M$, we use a neural network $g_2(h_t)$ to learn the mean and standard deviation $\{\mu_x, \sigma_x\}$, latent feature $z \in Z = \{z_1, z_2, \dots, z_n\}$, $Z \in \mathbb{R}^{n \times L}$, where L is the length of latent feature matrix Z .

In order to distinguish the normal latent features of different data attributes, we utilize the Gaussian distribution to model it as

$$p_\gamma(z|n) = \mathcal{N}(\mu_n, \text{diag}(\sigma_n^2)), \quad (14)$$

where n is the number of data attributes. It follows a multinomial distribution as:

$$p_\gamma(n) = \text{Mult}(\pi), \quad (15)$$

where π is the parameter of the multinomial distribution.

According to the research in [35], we can consider $\mu_n \in \mathbb{R}^M$ as the normal encoded features of each attribute in the latent space. The encoded features corresponding to attributes $x_1 \sim x_n$ are $\mu_1 \sim \mu_n$, respectively, and are stored in the memory module, implemented by the matrix $M \in \mathbb{R}^{n \times L}$.

The memory module [36], [37] includes a memory representation to represent the encoded feature and a weight matrix obtained from the similarity of the memory items and the input z . To address attribute misplacement, we re-integrate the encoded features by querying the similarity with memory items. The specific operation is as follows. First, we calculate the similarity between the encoded feature z and all memory items to obtain the weight vector w . In this paper, we use cosine similarity

$$w_i = \frac{z \cdot \mu_i^\top}{\|z\| \|\mu_i\|}. \quad (16)$$

Then, the memory module outputs $\hat{z}$

$$\hat{z} = wM = \sum_{i=1}^n w_i \mu_i. \quad (17)$$

The $\hat{z}$ is the final latent feature obtained by the deep probabilistic adaptive memory network. We use latent features $\hat{z}$ to reconstruction x'' can be formalized as follows.

$$p_\theta(x''|\hat{z}) = \mathcal{N}(\mu_x, \text{diag}(\sigma_x^2)), \quad (18)$$

The latent feature $\hat{z} \in \hat{Z} = \{\hat{z}_1, \hat{z}_2, \dots, \hat{z}_n\}$, $\hat{Z} \in \mathbb{R}^{n \times L}$. The flattened output of $\hat{Z}$, denoted as Z_m , and the flattened output of the self attention-based multi-scale convolutional network, denoted as Z_c , constitute a new latent space.

When concept drift occurs, it means that the statistical properties or distribution of the data have changed, which may render the information stored in the memory module no longer applicable to the current new data patterns. Therefore, we need to update the information in memory. An intuition is that we can assess whether concept drift has occurred by evaluating the similarity between z and the memory items stored in memory. If the similarity decreases, this indicates that the memory items in memory may no longer represent the current data distribution.

Definition 2(Memory similarity) For input data x , we can get the similarity between its latent features z and each memory item as w_i . The similarity between x and memory is defined as follows:

$$W_{sim} = \sum_{i=1}^n w_i. \quad (19)$$

Memory similarity may be negative. A negative memory similarity indicates the presence of unusual patterns in the data. This will attract the attention of the model and facilitate the detection of anomalies. To prevent the introduction of anomalies during memory updates, we have adopted the following two measures. First, we construct a comprehensive scoring mechanism that combines the similarity between the data and the memory module with the reconstruction error of the data. This mechanism provides a more accurate assessment of when memory updates are necessary (details are described in Subection 4.4). Second, when updating the memory module, we select an appropriate window length to ensure that sufficient new data are available to accurately capture the current data distribution (details are discussed in Section 5). We can more effectively identify concept drift and update the memory module accordingly by these measures. This ensures that the model can adapt to the evolution of the data, thereby enhancing the accuracy and robustness of anomaly detection.

C. Joint Optimization Objective

The objective of deep probabilistic adaptive memory network is to maximize the marginal log-likelihood of the training data and the formula is as follow.

$$\log p_\theta(X_1, X_2, \dots, X_N) = \sum_{i=1}^N \log p_\theta(X_i), \quad (20)$$

where $X_1, X_2, \dots, X_N$ are N training attribute samples, the evidence lower bound on the marginal likelihood of each X is

$$\begin{aligned} \log p_\theta(X) &\geq \mathbb{E}_{q_\phi(z|X)} [\log p_\theta(X|z)] \\ &\quad - \mathbb{E}_{q_\phi(z|X)} [D_{KL}(q_\phi(c|X) || p_\gamma(c))] \\ &\quad - \mathbb{E}_{q_\phi(c|X)} [D_{KL}(q_\phi(z|X) || p_\gamma(z|c))], \end{aligned} \quad (21)$$

where $D_{KL}(\cdot || \cdot)$ is the KL-divergence between two distributions.

After the model reconstructs all missing values, we measure the reconstruction error using the Mean Absolute Error (MAE). The reconstruction error loss consists of two parts. One part is the reconstruction error of the non-missing data

under the influence of the mask M , denoted as L_M . The other part is the reconstruction error of the artificially missing data under the influence of the artificial missing mask I , denoted as L_I . The reconstruction error loss is calculated as follows:

$$L_M = \frac{\sum_{i=1}^n \sum_{j=1}^T |(X_{ij} - X'_{ij}) \odot M_{ij}|}{\sum_{i=1}^n \sum_{j=1}^T M_{ij}}, \quad (22)$$

$$L_I = \frac{\sum_{i=1}^n \sum_{j=1}^T |(X_{ij} - X'_{ij}) \odot I_{ij}|}{\sum_{i=1}^n \sum_{j=1}^T I_{ij}}, \quad (23)$$

$$loss = \frac{1}{2}(L_M + L_I). \quad (24)$$

We adopt the standard supervised learning paradigm to train the model, optimizing the network parameters by minimizing the mean squared error loss between the predicted outputs and ground-truth results. Based on the training dataset and the loss function formulations in equations (22)-(24), we iteratively refine the model parameters via backpropagation. The training process terminates upon reaching the predefined number of epochs.

D. Anomaly Detection of Adaptive Threshold

After the model training is completed, we use our proposed model to perform anomaly detection on newly acquired data. Given a multivariate time series X that needs to be detected, obtain its reconstructed representation X' . The anomaly detection score $s(X)$ is calculated as follows.

$$s = \frac{|X_{\cdot j} - X'_{\cdot j}|}{n \sum_{sim=1}^n W_{sim}}. \quad (25)$$

where $X_{\cdot j}$ represents all elements of the j^{th} column of X . The absolute average error between the j^{th} column of X and the j^{th} column of X' is calculated. When this score s exceeds a given threshold, we consider the data at this moment to be anomalous.

In practice, the distribution of time-series data may change over time due to concept drift, making it challenging to detect anomalies using a single, static threshold. If the parameters are not set accurately, this may lead to a large number of false positives or false negatives. Therefore, we propose an adaptive threshold algorithm based on Extreme Value Theory [17].

Extreme value theory refers to the distribution of extreme events that we may observe by inferring without any distribution assumption based on the original data, which is the extreme value distribution (EVD). Its mathematical expression is as follows:

$$G_\gamma : x \mapsto \exp(-(1 + \gamma x)^{-\frac{1}{\gamma}}), \gamma \in \mathbb{R}, 1 + \gamma x > 1. \quad (26)$$

The γ is the extreme value index, which depends on the extreme value data of different distributions. In order to fit the EVD to the tail of the unknown input distribution, it is necessary to estimate γ . But it is difficult to calculate it effectively. To solve this problem, the Peaks-Over-Threshold (POT) approach is proposed [17]. This method depends on the

Pickands-Balkema-de Haan theorem [40] (also called second theorem in EVT) given below.

$$\bar{F}_t(x) = P(X - t > x | X > t) \sim (1 + \frac{\gamma x}{\sigma})^{-\frac{1}{\gamma}} \quad (27)$$

The result shows that the distribution of threshold exceeding part $(X - t)$ meets the requirements of Generalized Pareto Distribution(GPD) with parameters γ, σ . That is, the tail distribution can be fitted by the generalized Pareto distribution(GPD). γ and σ can be quickly calculated using the Grimshaw algorithm. From this, we can get the threshold value of the overall distribution, and the calculation formula is as follows:

$$z_q \simeq t + \frac{\hat{\sigma}}{\hat{\gamma}} ((\frac{qn}{N_t})^{-\hat{\gamma}} - 1) \quad (28)$$

where t is the initial threshold and the empirical value is 98%; σ and γ are parameters of GPD, which are obtained by maximum likelihood; q is the expected probability; N is the total number of current samples; N_t is the peak number, i.e. $X_i > t$. The POT strategy provides us with a method to estimate z_q . With the support of the aforementioned methods, the adaptive threshold approach is detailed in Algorithm 1.

According to the algorithm we proposed and Equation (25), the anomaly score $s(X)$ can be obtained. Based on Equations (26) to (28) and the anomaly score $s(X)$, we employ the POT strategy following the procedure outlined in Algorithm 1 to compute the anomaly threshold and produce the output. The key aspect of the POT strategy lies in calculating z_q using Equation (28) and determining anomalies by comparing the anomaly score $s(X)$ with z_q . Values of $s(X)$ that exceed z_q are placed into the anomaly set and subsequently output.

Algorithm 1: ADAT(Anomaly Detection of Adaptive Threshold)

Input: $s_1, \dots, s_n, n, q$

Output: Set of anomalies A

```

1  $A \leftarrow \emptyset, Y_t \leftarrow \emptyset, z_q, t \leftarrow \text{POT}(s_1, \dots, s_n, q);$ 
2  $k \leftarrow n;$ 
3 for  $i > n$  do
4   if  $s_i > z_q$  then
5     Add  $(i, s_i)$  in  $A$ ;
6   else if  $s_i > t$  then
7      $Y_i \leftarrow s_i - t;$ 
8     Add  $Y_i$  in  $Y_t$ ;
9      $N_t \leftarrow N_t + 1;$ 
10     $k \leftarrow k + 1;$ 
11     $\hat{\gamma}, \hat{\sigma} \leftarrow \text{Grimshawk}(Y_t);$ 
12     $z_q \leftarrow \text{CalcThreshold}(q, \hat{\gamma}, \hat{\sigma}, k, N_t, t);$ 
13    Update Memory;
14  else
15     $k \leftarrow k + 1;$ 
```

To update memory, we first measure the gap between data and memory within the window. Let W_{curr} be the sets of similarity scores on the current window and W_{last} be those on the last window used to update the model. To measure the difference between W_{last} and W_{curr} , we use the mean

difference bound method based on the Hoeffding inequality [48] to define reliability.

Theorem 1 (Hoeffding's Inequality-Based Mean Difference Bound): Given independent random variables X and Y bounded by $[a_{\min}, a_{\max}]$, the probability of the sample mean difference between $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{Y} = \frac{1}{m} \sum_{j=1}^m Y_j$ is bounded by

$$\Pr\{|\bar{X} - \bar{Y}| \geq \epsilon\} \leq e^{\frac{-2\epsilon^2}{(n^{-1} + m^{-1})(a_{\max} - a_{\min})^2}}. \quad (29)$$

We calculate memory reliability based on Theorem 1. The memory reliability within the current window is denoted as r_M and calculated as follows.

$$r_M = e^{\frac{-b\epsilon^2}{(q_{\max} - q_{\min})^2}}, \quad (30)$$

where

$$\begin{aligned} b &= |W_{\text{curr}}| = |W_{\text{last}}|, \\ \epsilon &= |\text{avg}(W_{\text{curr}}) - \text{avg}(W_{\text{last}})|, \\ q_{\max} &= \max(\max(W_{\text{curr}}), \max(W_{\text{last}})), \\ q_{\min} &= \min(\min(W_{\text{curr}}), \min(W_{\text{last}})). \end{aligned} \quad (31)$$

If r_M is less than the set threshold, the memory will perform incremental updates based on the potential feature representations obtained from the data information in the current window.

V. ANALYSIS

In this section, we investigate the optimal window size for effectively handling concept drift. Theoretically, a larger window can more accurately reflect the overall distribution of the data. However, an excessively large window may impede the model's ability to rapidly adapt to changes in the data distribution. Therefore, the key is to find a proper balance, ensuring that the window collects a sufficient amount of data in the shortest possible time to accurately understand the current data distribution. Such a window size should not only be representative of the overall data distribution but also be capable of swiftly responding to any new changes in the data distribution.

Our primary approach is to partition the data within the window into two segments with equal sample sizes. We then calculate the deviations of the sample means of the left segment, the right segment, and the entire window from the expected mean. If these sample means are close to the expected mean with high probability, it indicates that the window size is appropriate.

A window of length W is partitioned into two parts and the sample means of these two parts are $\bar{X}_1$ and $\bar{X}_2$, respectively. The mean of all samples within the window is denoted as $\bar{X}_w$. According to Hoeffding's inequality, we have

$$\Pr(\bar{X} - \mu \geq \epsilon) \leq \exp\left(-\frac{2n^2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right), \quad (32)$$

where n denotes the number of samples within the window. b_i and a_i represent the maximum and minimum values of the samples, respectively. The number of samples within the

window is equal to the window size, that is, $n = W$. We denote $S = (b_i - a_i)^2$. We have

$$A : \Pr(\bar{X}_1 - \mu_1 \geq \epsilon_1) \leq \exp\left(-\frac{W\epsilon_1^2}{S}\right) \quad (33)$$

$$B : \Pr(\bar{X}_2 - \mu_2 \geq \epsilon_2) \leq \exp\left(-\frac{W\epsilon_2^2}{S}\right) \quad (34)$$

$$\Pr(\bar{X}_w - \mu \geq \epsilon) \leq \exp\left(-\frac{2W\epsilon^2}{S}\right) \quad (35)$$

Obviously, we have $\bar{X}_w = (\bar{X}_1 + \bar{X}_2)/2$. Let the probability of event A occurring $\exp\left(-\frac{W\epsilon_1^2}{S}\right)$ be denoted as q_1 , and the probability of event B occurring $\exp\left(-\frac{W\epsilon_2^2}{S}\right)$ be denoted as q_2 . The probability of both events A and B occurring simultaneously $\Pr(AB)$ is given by

$$\begin{aligned} \Pr(AB) &= 1 - \Pr(\bar{A}) - \Pr(\bar{B}) \\ &= 1 - (1 - q_1) - (1 - q_2) \\ &= q_1 + q_2 - 1. \end{aligned} \quad (36)$$

We choose reasonable values for ϵ_1 and ϵ_2 such that q_1 and q_2 are greater than 0.5, thereby ensuring that the sample means $\bar{X}_1, \bar{X}_2$, and the population means μ, μ_2 are likely to be close with high probability.

Let $\epsilon = (\epsilon_1 + \epsilon_2)/2$ and $\epsilon_1 < \epsilon_2$. We have $0 < \epsilon_1 < \epsilon < \epsilon_2$. Let $D = \exp\left(-\frac{W\epsilon_1^2}{S}\right) + \exp\left(-\frac{W\epsilon_2^2}{S}\right)$. Owing to $\exp\left(-\frac{W\epsilon_1^2}{S}\right), \exp\left(-\frac{W\epsilon_2^2}{S}\right) \in (0.5, 1)$, we have $D \in (1, 2)$

If $(\mu_1 + \mu_2)/2 \geq \mu$, we have

$$\exp\left(-\frac{W\epsilon_1^2}{S}\right) + \exp\left(-\frac{W\epsilon_2^2}{S}\right) - 1 \leq \exp\left(-\frac{2W\epsilon^2}{S}\right). \quad (37)$$

The inequality (37) can be rewritten as follows,

$$D - 1 \leq \exp\left(-\frac{2W\epsilon^2}{S}\right). \quad (38)$$

We have

$$W \leq \frac{S \ln(D_{\min} - 1)}{-2\epsilon^2}. \quad (39)$$

If $(\mu_1 + \mu_2)/2 \leq \mu$, we have

$$\exp\left(-\frac{W\epsilon_1^2}{S}\right) + \exp\left(-\frac{W\epsilon_2^2}{S}\right) - 1 \geq \exp\left(-\frac{W\epsilon^2}{S}\right). \quad (40)$$

The inequality (40) can be rewritten as follows,

$$D - 1 \geq \exp\left(-\frac{2W\epsilon^2}{S}\right). \quad (41)$$

We have

$$W \geq \frac{S \ln(D_{\max} - 1)}{-2\epsilon^2}. \quad (42)$$

Inequalities (39) and (42) demonstrate that when the value of W is within the interval $[\frac{S \ln(D_{\max} - 1)}{-2\epsilon^2}, \frac{S \ln(D_{\min} - 1)}{-2\epsilon^2}]$, the W with probability $\exp\left(-\frac{W\epsilon_1^2}{S}\right) + \exp\left(-\frac{W\epsilon_2^2}{S}\right) - 1$ is reasonable.

VI. EXPERIMENT RESULTS

A. Dataset

SWaT (Secure Water Treatment) Dataset: This dataset originates from a water treatment experimental platform. Data is collected at a frequency of once per second over a period of 11 days, encompassing records from 51 sensors. The dataset is divided into a training set and a testing set, with the first 7 days serving as the training set, which does not contain anomalies or attack behaviors; the last 4 days serve as the testing set, which includes 36 attacks, with approximately 11% of the time steps labeled as anomalies.

WADI (Water Distribution) Dataset: This dataset simulates a water distribution system composed of a large number of water supply pipes, forming a more complete and realistic network for water treatment, storage, and distribution. The dataset includes 14 days of normal operation data, which constitute the training set. In the following days, several controlled physical attacks were conducted, which correspond to anomalies in the testing set, with attacks occurring at different time intervals.

SMD (Server Machine Dataset) Dataset: This dataset is collected from server machine data of a large internet company. The dataset is evenly divided into two parts, with the first half containing no anomalies and can be used as a training set; the second half contains attack behaviors and can be used as a testing set.

Regarding the above data, we will conduct experiments without any modifications for comparison and also experiments where data is altered to include position swapping and data missing for comparison.

Baselines: Several methods for detecting anomalies are used as baselines and be compared with our algorithm. An introduction to the baseline is shown below.

(1) **DAGMM [41]:** It is a Deep Autoencoding Gaussian Mixture Model (DAGMM) for unsupervised anomaly detection. The model utilizes a deep autoencoder to generate a low-dimensional representation and reconstruction error for each input data point, which is further fed into a Gaussian Mixture Model.

(2) **LSTM-VAE [42]:** The LSTM-VAE is a type of variational autoencoder that leverages the characteristics of long short-term memory networks. This model integrates signals and reconstructs their expected distribution by introducing a prior that varies with progress. When the detector finds that the anomaly score based on reconstruction exceeds the threshold set based on the state, it will report an anomaly.

(3) **GDN [43]:** Given high-dimensional time series data. This approach combines a structure learning approach with graph neural networks, additionally using attention weights to provide explainability for the detected anomalies.

(4) **USAD [44]:** It is based on adversely trained autoencoders. Its autoencoder architecture makes it capable of learning in an unsupervised way. The use of adversarial training and its architecture allows it to isolate anomalies while providing fast training.

(5) **TimesNet [45]:** To tackle the limitations of 1D time series in representation capability, TimesNet extends the analysis of temporal variations into the 2D space by transforming

the 1D time series into a set of 2D tensors based on multiple periods. This transformation can embed the intraperiod and interperiod-variations into the columns and rows of the 2D tensors respectively, making the 2D-variations to be easily modeled by 2D kernels.

(6) **DCdetector [46]:** DCdetector is a multi-scale dual attention contrastive representation learning model. DCdetector utilizes a novel dual attention asymmetric design to create the permuted environment and pure contrastive loss to guide the learning process, thus learning a permutation invariant representation with superior discrimination abilities.

(7) **FluxEV [10]:** FluxEV applies preprocessing on missing data and designs two interpolation strategies.

(8) **GST-Pro [5]:** GST-Pro proposed dynamic graph neural differential equations (DG-NCDEs) to model multivariate time series, effectively capturing the spatiotemporal features of the data even when it contains missing values.

(9) **AERO [47] :** AERO is a novel twostage framework tailored for unsupervised anomaly detection in astronomical observations. AERO is not only capable of distinguishing normal temporal patterns from potential anomalies but also effectively differentiating concurrent noise, thus decreasing the number of false alarms.

B. Metrics

We use precision, recall, and F1 score to represent the performance of our algorithm. F1 score is such an efficient method that many researchers use it as an evaluation criterion and we will use it as well. To calculate the F1 score, true positive (TP), false positive (FP), true negative (TN), and false negative (FN) are needed to calculate accuracy and recall. The formula is as follows,

$$precision = \frac{TP}{TP + FP}, \quad (43)$$

$$recall = \frac{TP}{TP + FN}, \quad (44)$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall}. \quad (45)$$

In all datasets, we currently adopt the point adjustment strategy that is commonly employed in the majority of existing literature. Although the point adjustment strategy has been widely adopted by many mainstream studies, to further rigorously validate the performance of our proposed method, we additionally report F1 scores computed without point adjustment (F1-w/o PA) for both our proposed method and baseline approaches. These results are presented in table III to table II.

Furthermore, in recent research works, several scholars have begun to adopt F1-PA%K [49], an evaluation metric that represents an intermediate compromise between the F1 point-adjusted strategy and the F1 non-point-adjusted strategy. Let $\hat{y}_i$ denote a data point to be detected. The computation for this

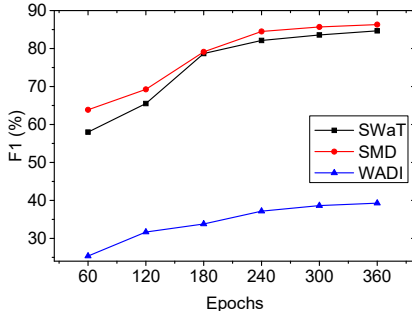


Fig. 2. The impact of varying epochs with 10% data missing rate

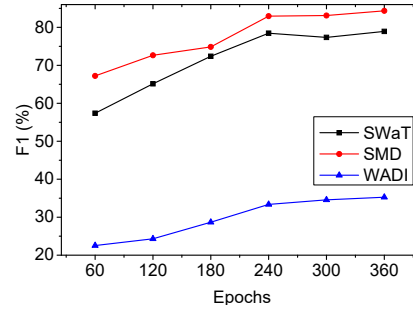


Fig. 3. The impact of varying epochs with 20% missing rate

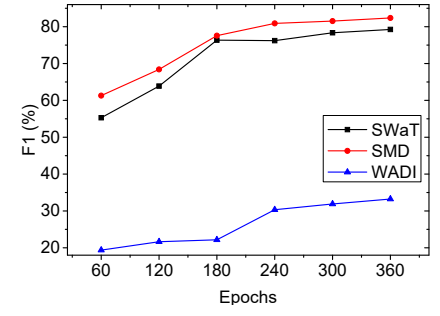


Fig. 4. The impact of varying epochs with 30% missing rate

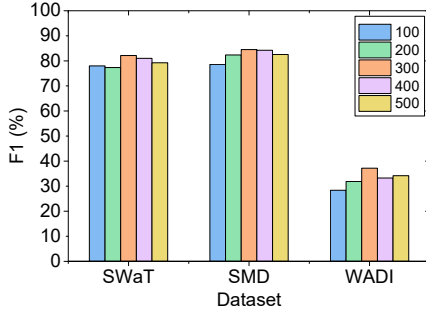


Fig. 5. The impact of altering sample length on the data with 10% missing rate

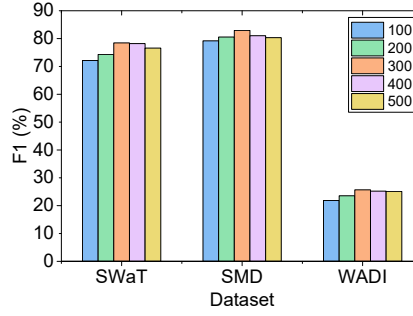


Fig. 6. The impact of altering sample length on the data with 20% missing rate

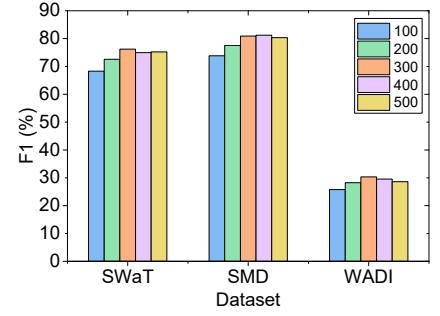


Fig. 7. The impact of altering sample length on the data with 30% missing rate

point being labeled as anomalous, i.e., $\hat{y}_t = 1$, is formulated as follows,

$$\hat{y}_t = \begin{cases} 1, & \text{if } \mathcal{A}(w_t) > \delta \text{ or} \\ & \left(t \in S_m \text{ and } \frac{|\{t' \mid t' \in S_m, \mathcal{A}(w_{t'}) > \delta\}|}{|S_m|} > K \right) \\ 0, & \text{otherwise,} \end{cases} \quad (46)$$

where $S_m = \{t_s^m, \dots, t_e^m\}$ denotes the m -th contiguous anomalous time segment, with t_s^m and t_e^m representing the start and end timestamps of this segment, respectively. $|S_m| = t_e^m - t_s^m + 1$ indicates the length of the anomaly segment; $\mathcal{A}(w_{t'})$ represents the anomaly score output by the model for time window $w_{t'}$; δ is the threshold employed for determining whether a single time point is anomalous; and $K \in [0, 100]$ serves as the core hyperparameter of the PA%K protocol, specifying the minimum detection ratio required to trigger the point adjustment mechanism. Specifically, point adjustment is applied to all time points within segment S_m only when the proportion of detected anomalous time points exceeds $K\%$. In this paper, the value of K is set to 30.

C. Performance Comparison

The original experimental data is complete without any missing values, attribute misplacement and concept drift. To meet the needs of our experiments, we make some missing values in some attributes of the original data. The types of missing values include single-point missing and continuous missing. The experiment designed three missing value scenarios to test the performance of anomaly detection algorithms after

data deletion under different missing rates. These scenarios include a missing rate of 10%, 20%, and 30%. Additionally, we swapped 2% of the data between two attributes of the original data and created a single instance of concept drift. In the following All experiments, we will adopt this setup. Considering that traditional multivariate time series anomaly detection models can only perform anomaly detection tasks on datasets without missing values, introducing missing values may render traditional models inexecutable. Therefore, before applying traditional models to datasets with missing values, it is necessary to first fill in the missing values. For different datasets, this study selected a value that is not within the reasonable range of these datasets, such as using -1000 to fill in missing values in a dataset with a value range of $[0, 100]$.

The comparison between our algorithm and the baseline methods is presented in table III to table II. The experimental results indicate that our algorithm performs better on low-quality data. This implies that our algorithm has better robustness.

To verify the superior performance of our algorithm in anomaly detection on missing data compared with other algorithms, we conducted comparisons with FluxEV, GST-Pro, USAD-mean and USAD-linear. FluxEV and GST-Pro are anomaly detection algorithms proposed to address data missing issues. USAD-mean and USAD-linear represent algorithms using mean value method and linear imputation respectively to fill in missing data before executing USAD algorithm. For the remaining baseline algorithms, we adopt the cubic spline imputation provided by GST-Pro. Ours-mean and Ours-linear represent our proposed method using mean

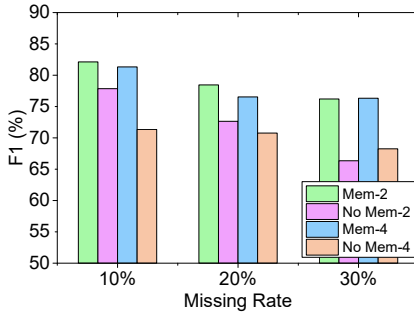


Fig. 8. The impact of memory network on the data with 10% missing rate

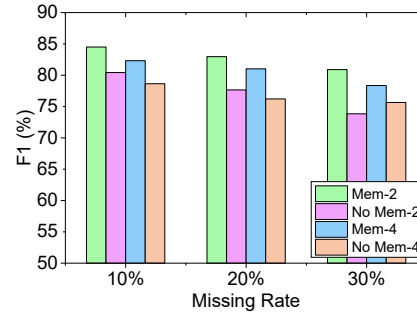


Fig. 9. The impact of memory network on the data with 20% missing rate

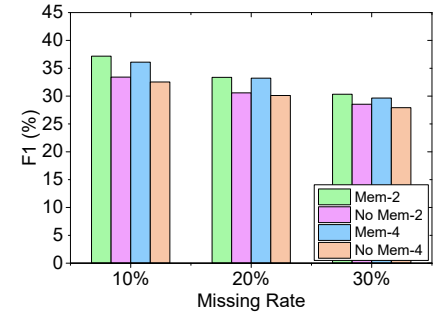


Fig. 10. The impact of memory network on the data with 30% missing rate

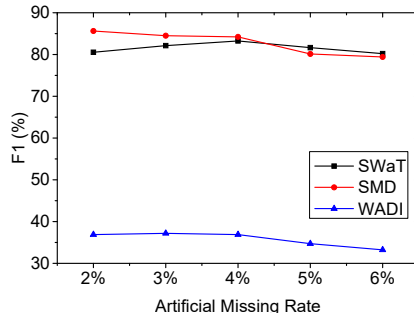


Fig. 11. The impact of artificial missing rate on the data with 10% missing rate

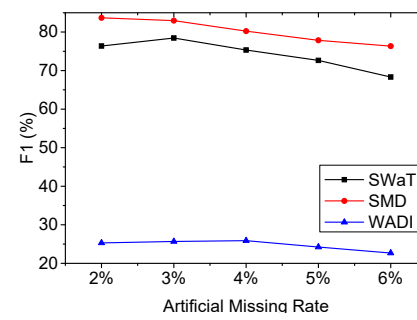


Fig. 12. The impact of artificial missing rate on the data with 20% missing rate

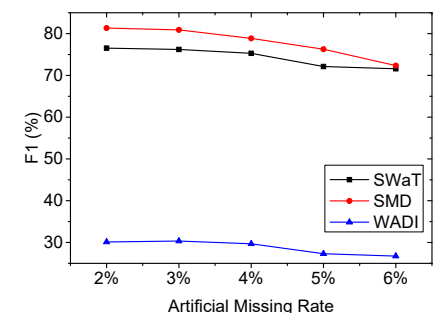


Fig. 13. The impact of artificial missing rate on the data with 30% missing rate

imputation and linear imputation to address missing data. The F1 results of Ours-mean, Ours-linear, and Ours in table III to table II indicate that simple use of mean and linear imputation may miss some anomalies. We use -1000, mask matrix $\mathbf{M}$ and missing length information $\mathbf{T}$ to better label missing data, rather than assuming that the missing data is a normal value like other imputation methods.

In table III to table II, our algorithm achieved overall higher F1, F1-w/o PA and F1-PA%K scores than other algorithms, indicating its stronger multivariate anomaly detection capability and better ability in handling complex missing data with situations like attribute misplacement and concept drift.

In scenarios that include a missing rate of 10%, 20%, 30%. and 2% data misplacement between two attributes of the original data and a single instance of concept drift, we investigate the impact of the number of epochs on the model's performance. From the experimental results in Fig.2 to Fig.4, we can observe that as the number of epochs increases, the model's accuracy also improves. When the model's number of epochs becomes excessively high, the rate of improvement in accuracy begins to slow down.

For different datasets, we set different training sample lengths to verify the impact of sample length on our experimental results. In low-quality datasets that include anomalies, concept drift, attribute misplacement, and data missing, the frequency and duration of anomalies are irregular. Sometimes, a larger sample length helps the model identify these anomalies, while sometimes the opposite is true. We conducted experiments on three datasets, simulating a data with different missing rate, and compared the impact of different sample

lengths such as 100, 200, 300, 400, and 500 on model performance. For example, in a dataset with S around 20, if we set the error $\epsilon = 0.07$, $D_{max} = 1.9$ and $D_{min} = 1.8$ then according to our optimal window size analysis, the most suitable window size is [215, 455].

The experimental results show that although there are cases where the model's F1 score is highest when the sample length is 400. Overall, however, the model's F1 score is highest and performs best when the sample length is 300. The specific comparative experimental results are shown in Fig.5 to Fig.7. In fact, a too short sample length is not conducive to model training, and a too long sample length will not improve training effectiveness.

D. Ablation Study and Parameter Analysis

We conducted an ablation study on the memory network to verify its function. The experimental results are displayed in Fig.8 to Fig.10. "Memo-2" indicates that the model has a memory network and 2 attributes are misplaced. "No Memo-2" indicates that the model does not have a memory network and 2 attributes are misplaced. "Memo-4" and "No Memo-4" have similar meanings. Without the use of a memory network, the occurrence of misplacement can lead to model misjudgments, thereby affecting the F1 score. What's worse, an increase in misplaced attributes can impact the model's performance and robustness.

To provide the model with reasonable feedback during training, we artificially make missing data in the low-quality data we set up. The proportion of this artificial missing data is an empirical value and we attempt to conduct a guided analysis

TABLE I
PERFORMANCE COMPARISON UNDER CONDITIONS OF 20% DATA MISSING, ATTRIBUTE MISPLACEMENT AND CONCEPT DRIFT

Dataset	SWaT			SMD			WADI		
Metric(%)	F1	F1-PA%K	F1-w/o PA	F1	F1-PA%K	F1-w/o PA	F1	F1-PA%K	F1-w/o PA
DAGMM	42.06	38.69	33.62	57.98	54.74	49.90	13.55	10.82	10.03
USAD	73.54	69.54	66.33	65.75	61.45	58.03	20.23	16.13	12.73
LSTM-VAE	59.34	55.57	50.75	77.88	74.11	69.32	22.19	18.26	13.49
GDN	68.13	64.74	59.34	53.55	50.21	45.10	14.63	9.04	1.97
TimesNet	69.61	66.10	61.67	76.85	73.34	68.96	28.69	24.30	18.62
DCdetector	71.80	68.03	63.81	80.91	77.15	73.01	29.65	25.06	20.01
AERO	71.85	68.57	63.47	69.88	66.60	61.95	25.09	21.04	15.94
FluxEV	71.08	67.26	62.91	76.59	72.92	68.44	28.91	24.79	18.16
GST-Pro	69.86	66.11	61.68	77.11	73.53	69.39	29.49	25.47	20.85
USAD-mean	71.41	67.73	63.40	66.53	62.90	57.59	21.05	17.31	12.98
USAD-linear	72.25	68.88	64.44	66.71	63.02	57.82	21.06	17.68	12.69
Ours-mean	76.07	73.41	70.79	80.96	77.07	75.76	31.94	29.20	26.26
Ours-linear	76.63	73.68	70.87	81.04	76.84	75.83	31.68	29.06	27.77
Ours	78.45	76.07	71.99	82.96	78.32	76.71	33.37	31.82	27.09

TABLE II
PERFORMANCE COMPARISON UNDER CONDITIONS OF 30% DATA MISSING, ATTRIBUTE MISPLACEMENT AND CONCEPT DRIFT

Dataset	SWaT			SMD			WADI		
Metric(%)	F1	F1-PA%K	F1-w/o PA	F1	F1-PA%K	F1-w/o PA	F1	F1-PA%K	F1-w/o PA
DAGMM	38.90	36.17	31.84	56.27	53.77	49.76	18.86	14.44	14.25
USAD	56.94	53.91	49.65	62.92	59.89	55.65	19.07	16.49	15.11
LSTM-VAE	58.62	56.06	52.07	74.95	72.44	68.42	28.09	25.32	23.41
GDN	66.95	64.71	60.39	39.74	37.38	32.48	12.45	8.47	1.82
TimesNet	68.57	65.78	61.91	74.49	71.82	68.04	25.78	22.92	18.56
DCdetector	71.80	68.81	63.90	78.76	76.21	72.36	27.31	24.29	20.01
AERO	69.64	66.98	63.07	67.57	65.04	61.26	23.26	20.38	15.71
FluxEV	68.77	66.09	62.23	74.51	71.89	68.08	26.09	23.14	18.16
GST-Pro	67.31	64.52	61.02	74.01	71.30	67.94	26.58	23.61	19.89
USAD-mean	68.29	65.84	61.61	64.25	61.90	57.00	20.20	17.64	13.35
USAD-linear	73.07	70.38	66.57	65.61	62.99	58.93	20.86	17.70	13.80
Ours-mean	74.14	71.31	69.36	78.72	75.92	74.11	27.85	24.75	23.17
Ours-linear	74.22	71.38	70.58	78.87	75.98	74.31	27.73	25.76	24.45
Ours	76.21	73.86	71.36	80.90	78.57	75.94	30.34	27.82	24.23

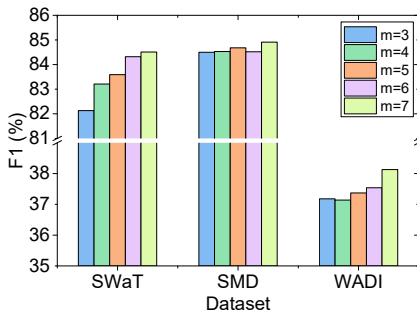


Fig. 14. The impact of the data with one attribute drift and varying memory number

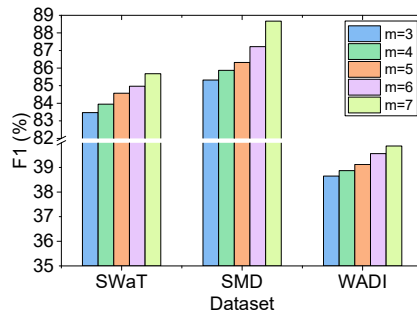


Fig. 15. The impact of the data with two attribute drift and varying memory number

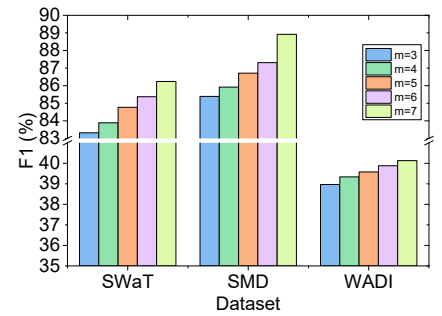


Fig. 16. The impact of the data with three attribute drift and varying memory number

TABLE III
PERFORMANCE COMPARISON UNDER CONDITIONS OF 10% DATA MISSING, ATTRIBUTE MISPLACEMENT AND CONCEPT DRIFT

Dataset	SWaT			SMD			WADI		
Metric(%)	F1	F1-PA%K	F1-w/o PA	F1	F1-PA%K	F1-w/o PA	F1	F1-PA%K	F1-w/o PA
DAGMM	49.54	42.41	39.85	64.34	57.31	54.84	12.13	11.69	11.46
USAD	74.03	65.72	64.20	69.81	61.61	59.28	21.78	20.54	19.11
LSTM-VAE	63.59	55.27	55.17	81.32	73.04	72.92	13.09	12.41	11.21
GDN	71.53	64.55	61.92	67.11	60.59	58.20	19.26	15.89	11.13
TimesNet	71.97	64.29	61.83	80.36	73.18	70.39	33.59	31.55	28.01
DCdetector	77.87	70.44	68.29	83.44	76.31	73.90	35.57	33.50	30.53
AERO	75.23	67.49	65.07	75.65	68.11	65.58	26.15	24.93	21.18
FluxEV	72.69	64.30	63.96	78.88	70.99	70.47	31.76	29.22	26.46
GST-Pro	71.96	64.58	62.58	78.02	71.14	68.99	32.63	30.14	27.30
USAD-mean	74.86	67.19	65.84	69.87	62.64	60.34	22.79	20.75	17.62
USAD-linear	74.09	66.17	64.44	72.58	65.48	62.67	22.97	20.91	18.53
Ours-mean	78.58	72.90	70.79	82.60	78.36	75.76	36.17	33.82	32.22
Ours-linear	79.23	72.95	71.07	81.93	78.55	74.93	35.73	33.46	31.81
Ours	82.13	75.84	74.13	84.50	80.60	76.71	37.18	34.89	34.23

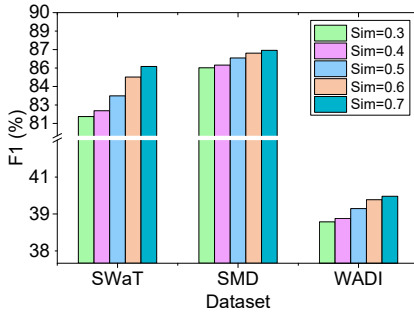


Fig. 17. The impact of model with 5 memories and varying similarity coefficient

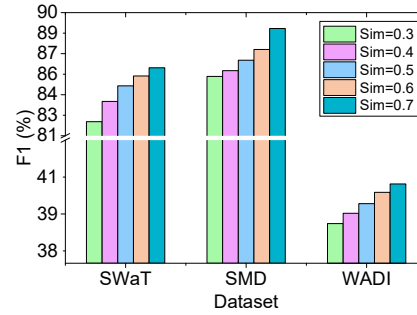


Fig. 18. The impact of model with 6 memories and varying similarity coefficient

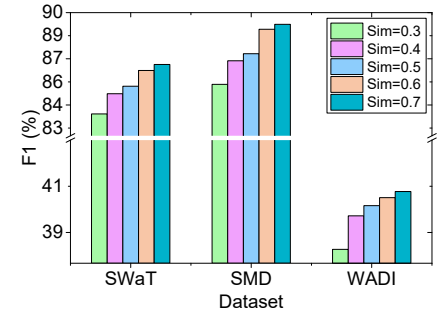


Fig. 19. The impact of model with 7 memories and varying similarity coefficient

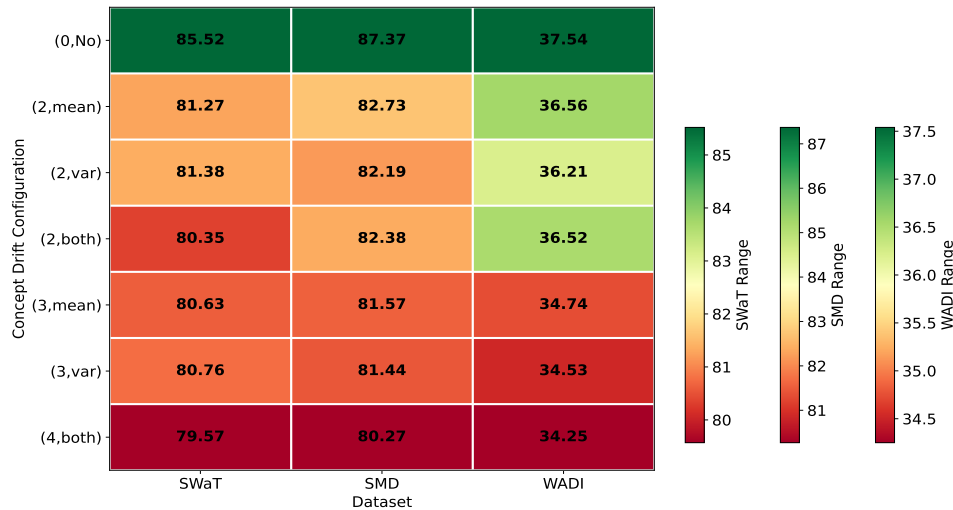


Fig. 20. Concept drift impact on anomaly detection performance.

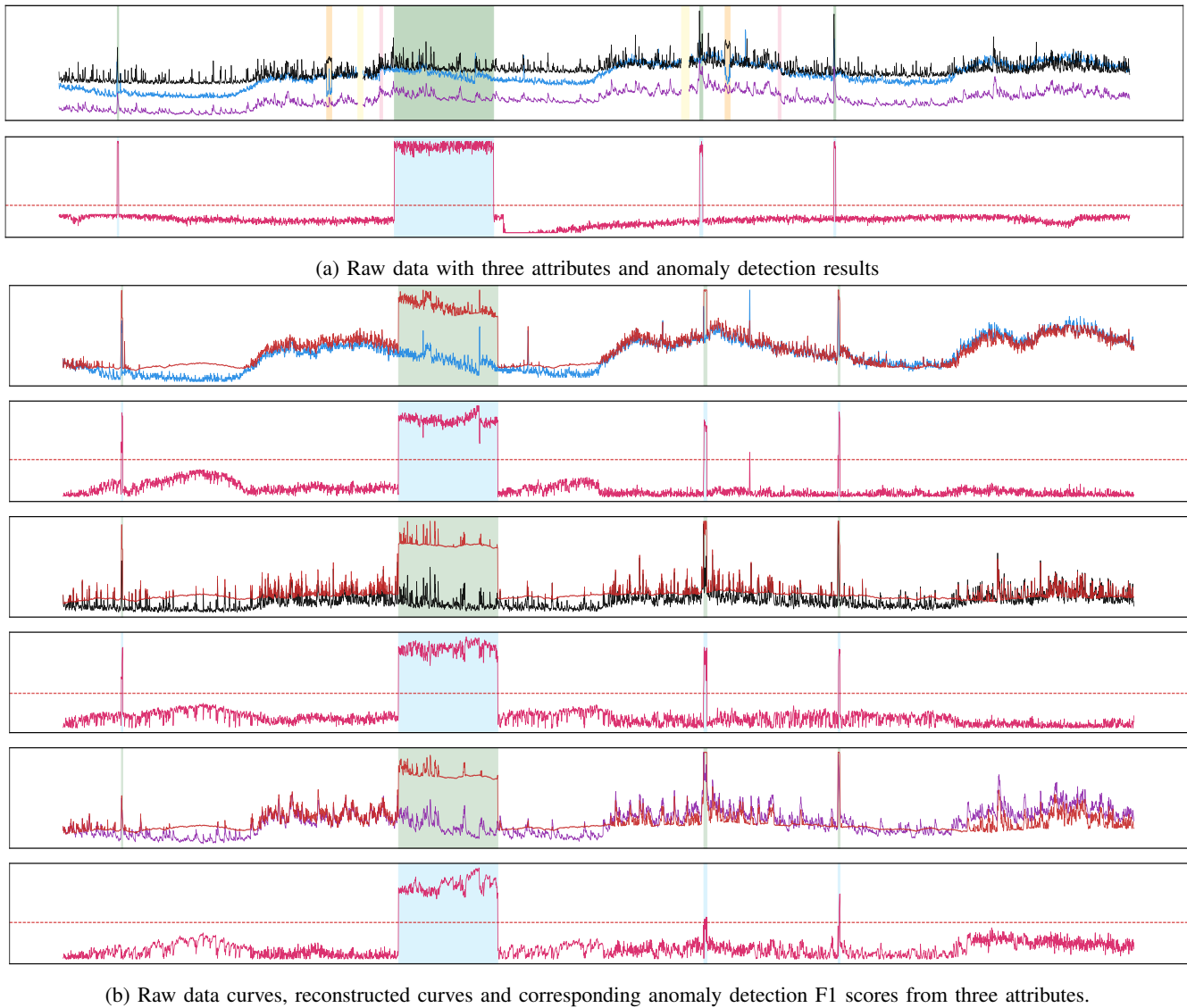


Fig. 21. Visualization of output from the proposed algorithm ADAT.

of it. The experiments are conducted on the low-quality dataset set up above, which includes anomalies, concept drift, attribute misplacement, and data missing. The data missing rates are still 10%, 20%, and 30%, respectively. Fig.11 to Fig.13 indicate that when the proportion of artificial missing data is low, such as 2% and 3%, the impact on the model's performance is limited. As the proportion of artificial missing data increases, the model's performance declines. The decline in model performance becomes more pronounced when the data itself has a higher missing rate. An increasing amount of missing data makes it more difficult for the model to accurately capture the relationships between attributes during the training phase.

We investigate the impact of the number of memories and the similarity coefficient on model performance. Additionally, we validate the model's capability in handling concept drift. The similarity threshold for the memories is set to 0.6. Under this configuration, we examine the effect of varying the number of memories from 3 to 7 on the F1 score. The experimental

results, depicted in Fig.14 to Fig.16, demonstrate that as the number of memories increases, the F1 score achieved by our algorithm also rises, indicating improved model performance. Furthermore, an increase in drifting attributes enhances the model's performance, as a greater number of drifting attributes more effectively substantiates the occurrence of concept drift, thereby reducing the false positive (FP) rate.

Since the number of memories significantly influences the model's performance, we fine-tuned the similarity coefficient threshold of the memories to regulate their quantity. We fixed the number of drifting attributes to 3. Under this data configuration, we investigated the effect of varying the similarity coefficient threshold from 0.3 to 0.7 on the F1 score. The experimental results are presented in Fig.17 to Fig.19. As the similarity coefficient threshold increased, the model's performance exhibited consistent improvement. A higher similarity coefficient threshold facilitates the merging of highly similar memories, ensuring that each memory retains distinct information. This optimization enhances the model's

ability to accurately detect anomalies in the data.

To verify the capability of our algorithm in handling concept drift, we designed an experiment incorporating multiple drifts and two types of drift scenarios. Based on a test set with a default configuration of 10% missing data and a misplacement ratio of 2% for two attributes, we introduced 2 to 4 drifts in two attributes of the test data, covering both mean and variance drift scenarios. The experimental results are shown in Fig.20. The X-axis represents the datasets, while the Y-axis denotes the configuration type and number of data drifts. The format for data drift configuration is (Number of concept drifts, Drift type). For example, (2, both) indicates that each attribute undergoes two concept drifts, including both mean drift and variance drift. The (0, No) indicates no drift occurrence.

The experimental results demonstrate that, compared to the test outcomes with only a single drift instance presented in table III to table II, an increase in the number of drifts leads to a certain degree of performance degradation in our algorithm. However, the rate of degradation slows down, and the extent of the decline remains limited. On the SWaT, SMD, and WADI datasets, the maximum performance degradation rates compared to scenarios without drift are 6.96%, 8.13%, and 8.76%, respectively. Furthermore, the impact of the two drift types on model performance varies across datasets, indicating that no single drift type consistently causes more significant performance deterioration.

E. Visual results analysis

In Fig.21, we visualize the prediction results of our algorithm on multivariate time series data containing both point anomalies and sequential anomalies. A segment with 4300 timestamps of the SMD test data including attribute 1, 2, and 3 is selected for visualization. In Fig.21(a), data issues such as data missing(yellow region), attribute misplacement(orange region), and concept drift(pink region) are each introduced twice in this segment. The green region indicates the ground-truth anomalies. Attribute misplacement occurs in attribute 1 and attribute 2. Other issues occur simultaneously in attribute 1, 2, and 3. We maintain the original values of other attribute data and use all attributes for model testing.

In Fig.21(b), the light blue region represents the detected anomalies. The blue, black, and purple raw data curves from three different attributes are each placed together with their corresponding red reconstructed curves. Below each attribute's raw data curve, the pink curve represents the anomaly score, and the dashed line indicates the anomaly threshold. Although we add different types of data issues to these three attributes in the raw data, the results show that our algorithm, by integrating data from other attributes, remains unaffected and successfully detects the true anomalies. In the anomaly detection of the third attribute, we do not find the first anomaly. However, after incorporating the detection results from the other attributes, this anomaly is ultimately not missed. Our model captures the spatiotemporal relationships across different attributes, rather than relying solely on a single or a few attributes.

VII. CONCLUSION

To tackle multivariate time series which includes datasets with missing data, attribute misplacement and concept drift, we propose an unsupervised anomaly detection framework. We propose a self-attention mechanism that incorporates masking and missing length information, thereby enhancing the model's capacity to manage incomplete data effectively. Besides, we propose a memory mechanism that leverages similarity-weighted items to obtain a reconstruction representation, correcting biases induced by attribute misplacement during the reconstruction process. Our experimental results demonstrate that our method outperforms other baseline algorithms in F1 Score when dealing with low-quality multivariate time series data.

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Algorithmica, IEEE Transactions on Knowledge and Data Engineering, SIGMOD, SIGKDD, VLDB, ICDE, and INFOCOM.